

PLANT DISEASE DETECTION USING CNN

*An Application Development – II (Project) Report Submitted
In partial fulfillment of the requirement for the award of the degree of*

*Bachelor of Technology
in
Computer Science and Engineering (Data Science)*

by

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MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

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CERTIFICATE

This is to certify that this is the bonafide record of the project titled **“Plant Disease Detection using CNN”** submitted by **CH.Nagasri(20N31A6711) A.Dheeraj(20N31A6703) and MD.Khaleel(21N35A6701)** of **B.Tech III Year – II Semester** in the partial fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)**, Dept. of CSE (Emerging Technologies) during the year 2022-2023. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

I hereby declare that the project entitled “**Plant Disease Detection using CNN**” submitted to **Malla Reddy College of Engineering and Technology**, affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH) as part of III Year B.Tech – II Semester and for the partial fulfillment of the requirement for the award of **Bachelor of Technology in Computer Science and Engineering (DataScience)** is a result of original research work done by us.

It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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ABSTRACT

- Early Disease Detection is important for better yield and quality of crops. With Reduction in Quality of the agricultural Product, Disease Plant can lead to the huge Economic Losses to the Individual farmers. In country like India whose major Population is involved in Agriculture It is very important to find the disease at early stages. Faster and precise prediction of plant disease could help reducing the losses .Plants and crops that are infected by pests have an impact on the country's agricultural production. Usually, farmers keep a close eye on the plants in order to discover and identify diseases. Plant disease detection can be done by looking for a spot on the diseased plant's leaves. The goal of this project is to create a Disease Recognition Model that is supported by leaf image classification. To detect plant diseases, we are utilizing image processing with a Convolution neural network (CNN). A convolutional neural network (CNN) is a form of artificial neural network that is specifically intended to process pixel input and is used in image recognition.

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CHAPTER 1:

INTRODUCTION

Agricultural production is a very old means of obtaining food. It is a vital source of income for people all around the world. No one can exist in our world without food. Plants are crucial not only for humans, but also for animals who rely on them for food, oxygen, and other necessities. The government and experts are taking significant initiatives to enhance food production, and they are working successfully in the real world. When a plant becomes afflicted with a disease, all living organisms in the environment are affected in some way. This plant disease can affect anywhere on the plant, including the stem, leaf, and branch. Even the types of illnesses that impact plants, such as bacterial and fungal diseases .etc can differ. This type of natural tragedy is unavoidable. Early detection of plant disease aids in the prevention of large-scale crop losses. Farmers must apply the appropriate insecticides for their crops. Too many pesticides are harmful to crops and farmland. Getting expert advice will help you avoid misusing chemicals on plants. Plants have been the focus of many researchers to aid farmers and others involved in agriculture. When a disease is visible to the naked eye, it is straightforward to detect. The illness may be discovered and treated early if the farmer has sufficient information and monitors the crops on a regular basis. However, this phase only exists when the disease is extreme or crop output is low. Then there are the different innovations. Farmers will benefit from the introduction of automated disease detection tools. These technologies rely heavily on deep learning and neural network. Deep Convolutional Neural Network is utilized in this study to identify infected and healthy leaves, as well as to detect illness in afflicted plants.

1.1 PROBLEM DEFINITION

Plant diseases have turned into a dilemma as it can cause significant reduction in both quality and quantity of agricultural products. Automatic detection of plant diseases is an essential research topic as it may prove benefits in monitoring large fields of crops, and thus automatically detect the symptoms of diseases as soon as they appear on plant leaves. The proposed system is a software solution for automatic detection and classification of plant leaf diseases. The scheme consists of four main steps, first a color transformation structure for the input RGB image is created, then the green pixels are masked and removed using specific threshold value followed by segmentation process, the texture statistics are computed for the useful segments, finally the extracted features are passed through the classifier.

1.2: EXISTING SYSTEM

- The Existing System for plant disease detection is simply naked eye observation by experts through which identification and detection of plant diseases.
- Traditional approaches were used for predicting the plant diseases.
- Existing System has lower accuracy to predict the plant diseases.

1.3: PROPOSED SYSTEM

- The proposed system begin with the thought that was not executed by the ancestors.
- Image pre-processing is an improvement of the image data that suppresses unwanted distortions or enhances some image features important for further processing.
- If it is image dataset then we can make array of images and then apply model like CNN depending upon how much accuracy it gives.
- Plant disease prediction can be done using “Crop Disease Dataset”.

CHAPTER 2: SYSTEM REQUIREMENTS

Requirement analysis is involved in defining customer needs and objectives in the context of planned customer use, and system features identified to determine the needs of system functions.

User Requirements: -

User engagement with factual statements and assumptions that define system expectations based on objectives, environment, issues and measures of efficiency and effectiveness.

Basically, users need:

- i) A system that improves the efficiency of data retention and retrieval.
- ii) An easy-to-read and use system.
- iii) A system that speeds up the processing of transactions.
- iv) A flexible, secure and flexible system.

2.1: SOFTWARE REQUIREMENTS

- Platform : python
- Algorithm : CNN, Gaussian Naive Bayes
- Tool :python 3.8
- OS : windows 7 or higher

Languages used: -

Python

- Python is a high-level, general purpose computer language. Python can be used on a server to create web applications. Python works on different platforms (Windows, Mac, Linux, Raspberry pi,etc).Python runs on a interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick

Pandas

- Pandas is an open source Python package that is most widely used for data science/data analysis and machine learning tasks. It is built on top of another package named Numpy, which provides support for multi-dimensional arrays. Pandas is a Python library used for working with data sets. It has functions for analyzing, cleaning, exploring, and manipulating data.

Matplotlib

- Matplotlib is a cross-platform, data visualization and graphical plotting library(histograms,scatterplots,barcharts....etc) for Python and its numerical extension numpy.

Keras

- Keras is used for creating deep models which can be productized on smartphones. Keras is also used for distributed training of deep learning models. Keras is used by companies such as Netflix, Yelp, Uber, etc.

Tensorflow

- TensorFlow allows you to create dataflow graphs that describe how data moves through a graph. The graph consists of nodes that represent a mathematical operation. A connection or edge between nodes is a multidimensional data array.

Numpy

- NumPy can be used to perform a wide variety of mathematical operations on arrays. It adds powerful data structures to Python that guarantee efficient calculations with arrays and matrices and it supplies an enormous library of high-level mathematical functions that operate on these arrays and matrices.

2.2 HARDWARE REQUIRMENTS

1. Ram: 4GB or more.
2. Processor: Intel Core Dual
3. Programming Language: Python 3.6.
4. Hard disk space: 10GB or more.

CHAPTER 3

SYSTEM DESIGN

System Architecture Diagram:

To diagnose the reason for the symptom by using an automatic tool, therefore the image processing system is proposed to develop to automate the identification and classification of the leaf batches into specific disorders. As shown within the figure above the system consists of three main blocks: Image Analyzer, Feature Database and Classifier resp. [9]. The processing proposed to try to by these blocks is split into two phases as follows offline Phase: an outsized set of defected images are processed by a picture analyzer for extracting abnormal features.

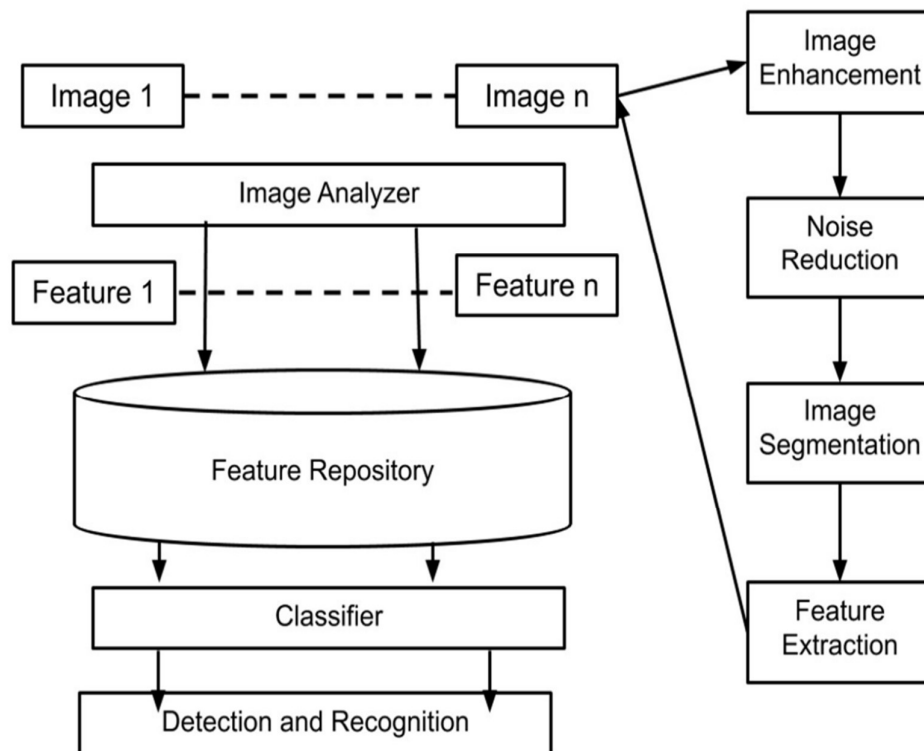


Fig 1. Architecture

Flow Chart:

The input test image is developed and pre-processed in the following phase and then it is transformed into array form for difference. The chosen database is appropriately separated and pre-processed and then retitled into suitable folders. The model is well trained using CNN and then classification takes position. The evaluation of the test image and the trained model take position tracked by the display of the result. If there is a flaw or infection in the plant the package displays the disease along with the remedy.

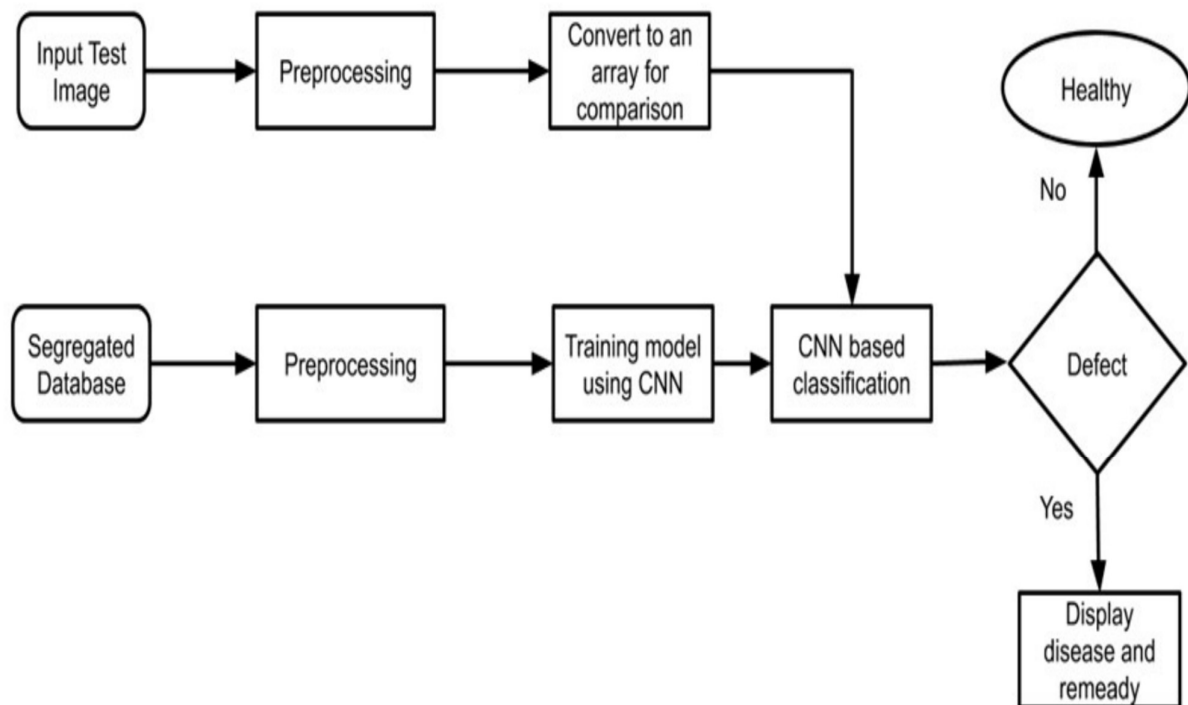
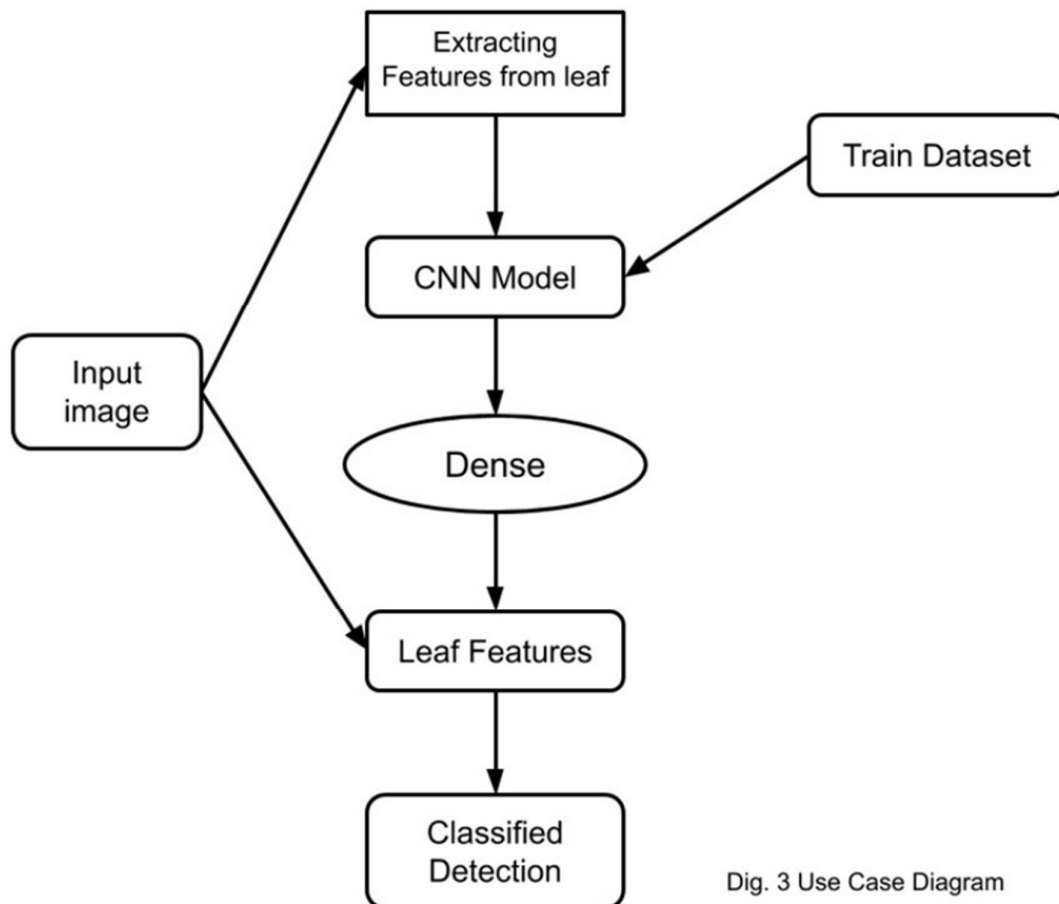


Fig 2.Data Flow Diagram

Use Case Diagram:

When we provide a new input image first the module extracts the leaf features. Then it goes through the CNN model. It then compares the features with already trained dataset. Then it goes through dense CNN and therefore the leaf features are extracted separately. Then the module will predict whether the plant leaf is affected by any disease or not. It shows the output from one among the 38 classes which are predetermined and trained. Then the output is going to be during a textual format.



Dig. 3 Use Case Diagram

Fig 3. User Case Diagram

Class Diagram:

The normalization class comprises raw images and it is fed to the CNN model which comprises dense and weight. The CNN model categorizes and identifies by using the training model. The training model class contains the image dataset. Leaf recognition becomes utilized of the features.

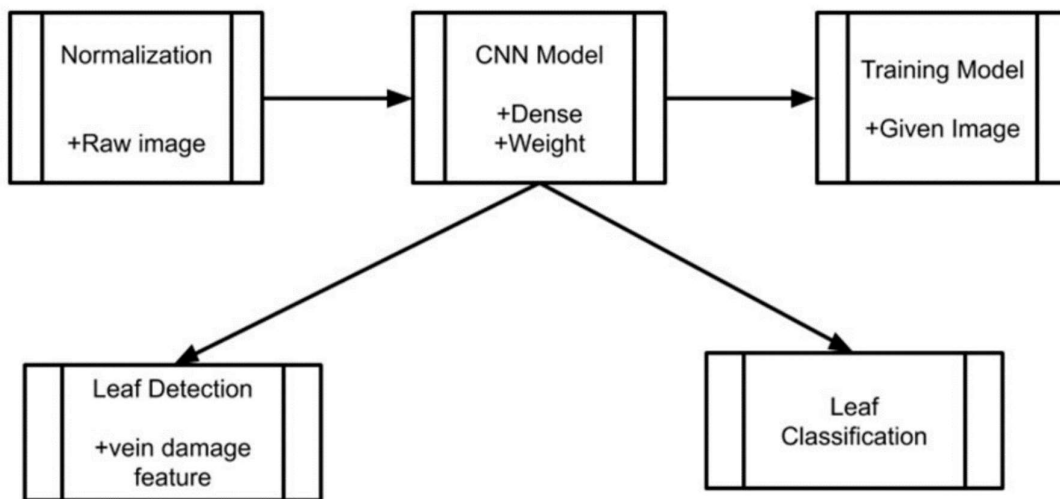


Fig 4: Class Diagram

3.2 MODULES

Module 1: Input module :

In this module, the capturing of image takes place. The image stored in this process is later used for detecting the disease . the input image will be store here

Module II: Image Processing :

In this module, the programs run over gives the image for further operation. When all the images are perfectly processed assign the program is ready for feature extraction

Module III: Extraction :

In this module, the programs has completed the Image Processing and is ready to extract the features from the given image . which is processed in before it will stores the features in it

Module IV:Classification:

In this module, the stored features of image will be classified with help of database which already existed . here classification of the features will be done on basis of color and shape.

Module V:Disease Prediction:

In this module, the classified features will be displayed and the prediction is done. Disease is predicted by comparing the features with available features in dataset.

4.IMPLEMENTATION

In our proposed work, we aim to detect and classify the plant diseases using the concept of Convolutional Neural Networks (CNN). Convolutional Neural Networks is a class of artificial neural network that works on the concept of hidden layers and is most widely used to in image processing and recognition. The basic framework or flow of our project consists of 5 steps, namely: Image Acquisition, Image Pre-processing, Image Segmentation, Feature Extraction and Classification.

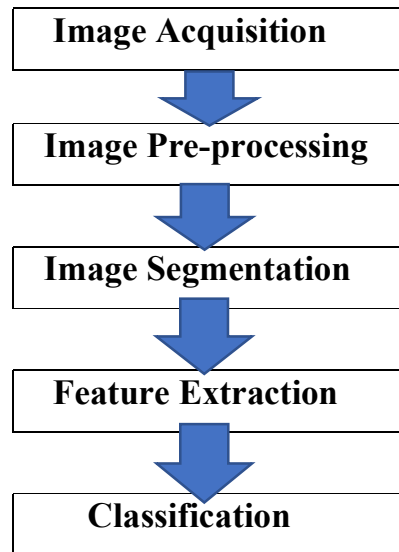


Image Acquisition

The first step of image classification is image acquisition. The dataset which is selected to build the model plays a very important role in the result observed from it. In imageoriented datasets, the images must be properly labelled, and it must be made sure that no such images exist which do not have labels. For our CNN model we have made use of PlantVillage dataset. This dataset has a collection of 15 directories of healthy and diseased plant leaves of 3 different plants namely Bell Pepper, Potato and Tomato. With a total of 20,639 images, this dataset houses the following directories of plant leaves

Image Pre-processing

The images which are acquired cannot be directly fed into the Convolutional Neural Network. Since, the computer does not understand anything other than binary numbers, the images also must be converted into machine understandable format. This is done by converting the images into array format. The images of the dataset are converted into 256 x 256 size arrays where each block represents the pixel value. The acquired images may have noise and other discolorations which are not ideal for image processing. All of these ambiguities have to be handled. This is carried out using the process of image segmentation, where the images are divided into smaller segments to map out the areas of interest and to avoid the areas which are less important, and which may cause disruptions to the accuracy of the system.

Feature Extraction

After the images are processed into proper format and divided into segments, the next very important step is feature extraction. This step in Convolutional Neural Network is done by using activation function followed by max-pooling layers.

Activation Function

The main purpose of using an activation function in a CNN is to decide which combination of weights and input will fire the next neuron. The activation function used in our CNN model is the ReLU activation function or Rectified Linear Unit activation function. This is very simple mathematical function, which outputs the input as long as the input is greater than zero, otherwise outputs zero. It can be represented as $y = \max(0, x)$ where x is the input and y is the output.

Max-Pooling Layers

In CNN, usually every activation function layer is followed by a pooling layer. Pooling layers are used to reduce the dimensions of feature maps. There are 3 types of pooling – average-pooling, max-pooling and global-pooling. We have made use of max-pooling technique because it enhances the features of the images by taking the maximum of the values present in that region. At the end of this operation, we obtain the most prominent features of the image. This step helps in avoiding over-fitting and also reduces noise distortion. Other than these 2 techniques used for feature extraction of an image, dropout layer is used to reduce over-fitting of the model, where the neurons are dropped at random. Flatten layer is used to convert the output into one long feature vector which is given as input into the next layer

Classification

After the layers of the Convolutional Neural Network are decided, we compile the model. The dataset is divided into training and test datasets, we have used 20% of the dataset for validation. We have an epoch value of 25, where epoch value is the number of passes that the model makes over the dataset. Real-time image processing functions are used to capture the different features of the image. Once, the model is trained, it is tested on the validation dataset

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.image import imread
import cv2
import random
import os
```

```

from os import listdir
from PIL import Image
from sklearn.preprocessing import label_binarize, LabelBinarizer
from keras.preprocessing import image
from keras.preprocessing.image import img_to_array, array_to_img
from tensorflow.keras.optimizers import Adam
from keras.models import Sequential
from keras.layers import Conv2D, MaxPooling2D
from keras.layers import Activation, Flatten, Dropout, Dense
from sklearn.model_selection import train_test_split
from keras.models import model_from_json
from tensorflow.keras.utils import to_categorical
plt.figure(figsize=(12,12))
path = "../input/leaf-image-dataset/Plant_images/Potato___Early_blight"
for i in range(1,17):
    plt.subplot(4,4,i)
    plt.tight_layout()
    rand_img = imread(path + '/' + random.choice(sorted(os.listdir(path))))
    plt.imshow(rand_img)
    plt.xlabel(rand_img.shape[1], fontsize = 10)#width of image
    plt.ylabel(rand_img.shape[0], fontsize = 10)
Converting Images to array
def convert_image_to_array(image_dir):
    try:
        image = cv2.imread(image_dir)
        if image is not None :
            image = cv2.resize(image, (256,256))
            #image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
            return img_to_array(image)

```

```

        else :
            return np.array([])
    except Exception as e:
        print(f"Error : {e}")
        return None

dir = "../input/leaf-image-dataset/Plant_images"
root_dir = listdir(dir)
image_list, label_list = [], []
all_labels = ['Corn-Common_rust', 'Potato-Early_blight', 'Tomato-
Bacterial_spot']
binary_labels = [0,1,2]
temp = -1

# Reading and converting image to numpy array
#Now we will convert all the images into numpy array.

for directory in root_dir:
    plant_image_list = listdir(f"{dir}/{directory}")
    temp += 1
    for files in plant_image_list:
        image_path = f"{dir}/{directory}/{files}"
        image_list.append(convert_image_to_array(image_path))
        label_list.append(binary_labels[temp])
label_counts = pd.DataFrame(label_list).value_counts()
label_counts.head()
image_list[0].shape
label_list = np.array(label_list)
label_list.shape

```

```

x_train, x_test, y_train, y_test = train_test_split(image_list, label_list,
test_size=0.2, random_state = 10)
x_train = np.array(x_train, dtype=np.float16) / 225.0
x_test = np.array(x_test, dtype=np.float16) / 225.0
x_train = x_train.reshape( -1, 256,256,3)
x_test = x_test.reshape( -1, 256,256,3)
y_train = to_categorical(y_train)
y_test = to_categorical(y_test)
model = Sequential()
model.add(Conv2D(32, (3, 3), padding="same",input_shape=(256,256,3),
activation="relu"))
model.add(MaxPooling2D(pool_size=(3, 3)))
model.add(Conv2D(16, (3, 3), padding="same", activation="relu"))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Flatten())
model.add(Dense(8, activation="relu"))
model.add(Dense(3, activation="softmax"))
model.summary()
model.compile(loss = 'categorical_crossentropy', optimizer =
Adam(0.0001),metrics=['accuracy'])
x_train, x_val, y_train, y_val = train_test_split(x_train, y_train, test_size =
0.2)epochs = 50
batch_size = 128
history = model.fit(x_train, y_train, batch_size = batch_size, epochs = epochs,
validation_data = (x_val, y_val))
plt.figure(figsize=(12, 5))
plt.plot(history.history['accuracy'], color='r')
plt.plot(history.history['val_accuracy'], color='b')
plt.title('Model Accuracy')

```

```
plt.ylabel('Accuracy')
plt.xlabel('Epochs')
plt.legend(['train', 'val'])
plt.show()
print("[INFO] Calculating model accuracy")
scores = model.evaluate(x_test, y_test)
print(f"Test Accuracy: {scores[1]*100}")
y_pred = model.predict(x_test)
img = array_to_img(x_test[10])
img
print("Originally : ",all_labels[np.argmax(y_test[10])])
print("Predicted : ",all_labels[np.argmax(y_pred[10])])
```

CNN Model Steps

Conv2D: It is a 2D Convolution Layer, this layer creates a convolution kernel that's wind with layers input which helps produce a tensor of outputs.

```
keras.layers.Conv2D(filters, kernel_size, strides=(1, 1),  
padding='valid',data_format=None, dilation_rate=(1, 1),  
activation=None, use_bias=True,  
kernel_initializer='glorot_uniform', bias_initializer='zeros',  
kernel_regularizer=None, bias_regularizer=None, activity_regularizer=None,  
kernel_constraint=None, bias_constraint=None
```

Maxpooling: Max pooling may be a pooling process that choose the very best element from the region of the feature map covered by the filter. Thus, the output after max-pooling level would be a feature map comprising the foremost important features of the previous feature map.[10].

Flatten: In between the convolutional layer and therefore the fully connected layer, there is a 'Flatten' layer. Flattening transforms a two-dimensional matrix of features into a vector which will be fed into a totally connected neural network classifier.

Image Data Generator: Image Data Generator quickly found out Python generators which will automatically turn image files on disk into batches of preprocessed tensors.

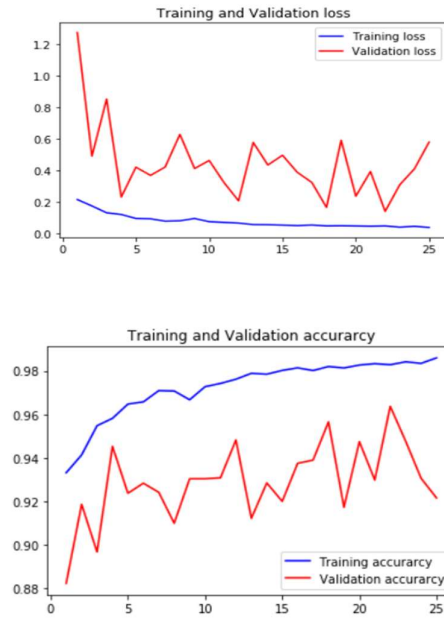
Training Process: Effective training begins well before a trainer delivers a private training session and continues then training session is complete. Training are often viewed as a process comprised of 5 related stages or activities: assessment, motivation, design, delivery, and evaluation.

Epochs: An epoch may be a term utilized in machine learning and indicates the amount of passes of the whole training dataset the machine learning algorithm has completed. Datasets are usually grouped into batches (especially when the quantity of knowledge is extremely large).

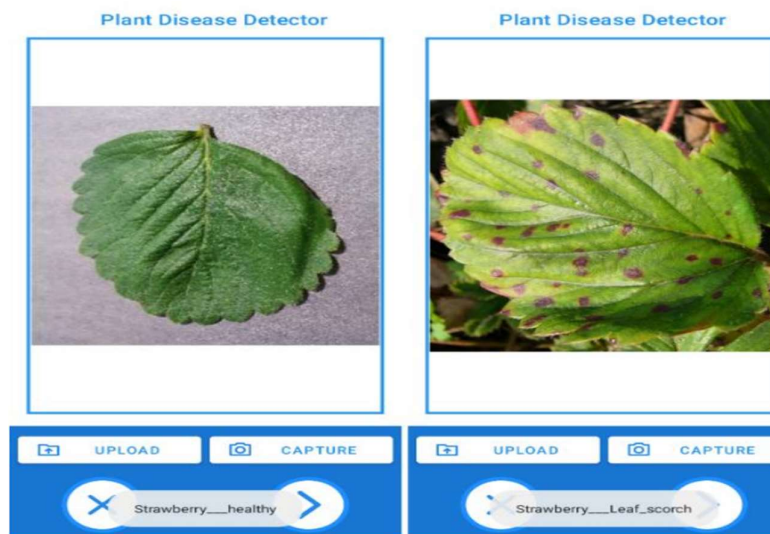
Validation Process: Validation is mentioned because the process where a trained model is evaluated with a testing data set. The testing data set may be a separate portion of an equivalent data set from which the training set springs . the most purpose of using the testing data set is to check the generalization ability of a trained model.

Training and Testing Model: The dataset is preprocessed like Image reshaping, resizing and conversion to an array form. Similar processing is additionally done on the test image. A dataset consisting of about 38 different plant leaf diseases is obtained, out of which any image is often used as a test image for the software. The train dataset is employed to coach the model (CNN) so that it can identify the test image and therefore the disease it is CNN has different layers that are Dense, Dropout, Activation, Flatten, Convolution2D, and maxpooling2d. After the model is trained successfully, the software can identify the disease if the plant species is contained within the dataset. After successful training and preprocessing, comparison of the test image and trained.

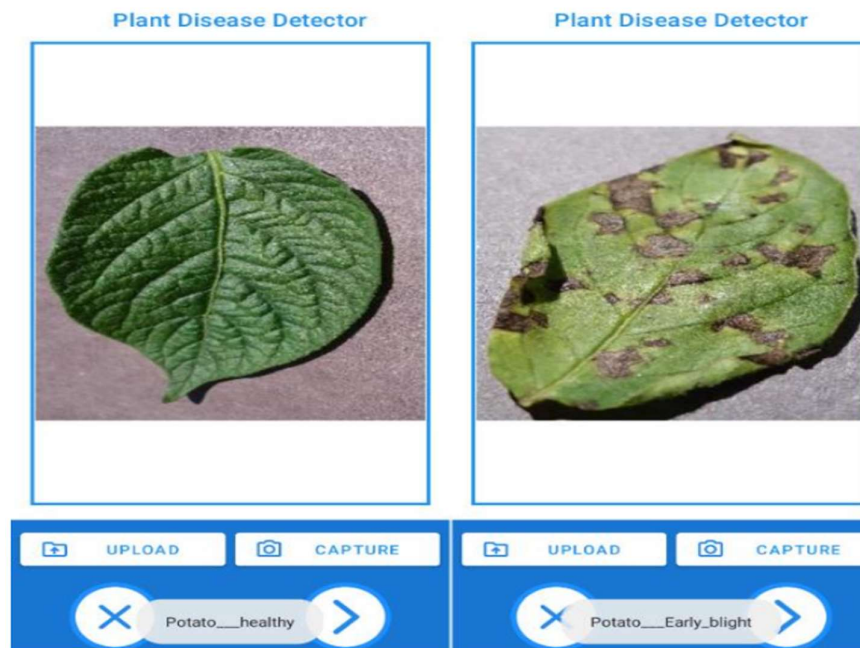
5.RESULTS:



Fig(a)



Fig(b)



Fig(c)

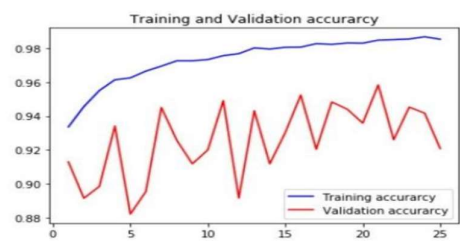


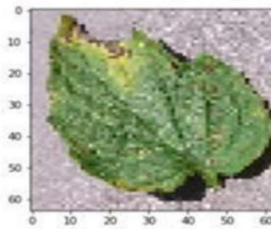
Fig -4: Plot of Training and Validation Accuracy



Fig(d)

SAMPLE OUTPUTS

```
In [50]: import matplotlib.pyplot as plt  
img = plt.imshow(test_image)
```



```
In [51]: test_image=image.img_to_array(test_image)
```

```
In [52]: test_image=np.expand_dims(test_image,axis=0)
```

```
In [53]: result=model.predict(test_image)
```

```
In [54]: prediction = result[0]
```

```
print("Tomato_Leaf_Mold")  
elif output['Tomato__Septoria_leaf_spot']==1.0:  
    print("Tomato_Septoria_leaf_spot")  
elif output['Tomato__Spider_mites Two-spotted_spider_mite']==1.0:  
    print("Tomato_Spider_mites_Two-spotted_spider_mite")  
elif output['Tomato__Target_Spot']==1.0:  
    print("Tomato_Target_Spot")  
elif output['Tomato__Tomato_Yellow_Leaf_Curl_Virus']==1.0:  
    print("Tomato_Tomato_Yellow_Leaf_Curl_Virus")  
elif output['Tomato__Tomato_mosaic_virus']==1.0:  
    print('Tomato_Tomato_mosaic_virus')  
elif output['Tomato__healthy']==1.0:  
    print("Tomato_healthy")
```

```
Tomato_Septoria_leaf_spot
```

6.CONCLUSION

- The proposed system describes different techniques of image processing for detecting leaf diseases.
- The disease of the plant is known at an early stage and the cure is suggested using different languages namely Hindi or English.
- As an extension to the project the number of classes of plants and its diseases will be increased .
- Also the model will be further improved by increasing the parameters for training and test.

FUTURESCOPE

- Anyone, literate or illiterate will be able to use easily this application.
- It will help Agriculture Officer to check quality of crop without any manual supervision.
- Data obtained from farmers will help Government to identify the disease produced in particular region and thus take proper action to prevent or control it.

7.REFERENCES

- [1] Diseased plant leaves using Neural Network Algorithms K. Muthukannan¹, P. Latha², R. Pon Selvi¹ and P. Nishal ¹Department of ECE, Einstein College of Engineering, Anna University, Tirunelveli, India ²Department of CSE, Government College of Engineering, Anna University, Tirunelveli, India ARPN Journal of Engineering and Applied Sciences, VOL. 10, NO. 4, MARCH 2015, ISSN 1819-6608
- [2] Ranjan, Malvika, and others "Detection and classification of leaf disease using artificial neural network." International Journal of Technical Research and Applications 3.3 (2015): 331-333.
- [3] Ishak, Syafiqah, and others. "Leaf disease classification using artificial neural network." Jurnal Teknologi 77.17 (2015).
- [4] Sladojevic, Srdjan, and others. "Deep neural networks based recognition of plant diseases by leaf image classification." Computational intelligence and neuroscience 2016 (2016).
- [5] Cortes, Emanuel. "Plant disease classification using convolutional networks and generative adversarial networks." (2017).
- [6] Walleign, Serawork, Mihai Polceanu, and Cedric Buche. "Soybean plant disease identification using convolutional neural network." The thirty-first international flairs conference. 2018.
- [7] Fuentes, Alvaro, and others. "A robust deep-learning based detector for real-time tomato plant diseases and pests recognition." Sensors 17.9(2017): 2022.
- [8] Liu, Peide, Tahir Mahmood, and Qaisar Khan. "Multiattribute decision making based on prioritized aggregation operator under hesitant intuitionistic fuzzy linguistic environment." Symmetry 9.11 (2017): 270.

[9] Mohanty, Sharada P., David P. Hughes, and Marcel Salath'e. "Using deep learning for image-based plant disease detection." *Frontiers in plant science* 7 (2016): 1419.

[10] Dhakal, Ashwin, and Subarna Shakya. "Image-based plant disease detection with deep learning." *International Journal of Computer Trends and Technology* 61.1 (2018): 26-29.

[11] Ramcharan, Amanda, et al. "Deep learning for image-based cassava disease detection." *Frontiers in plant science* 8 (2017): 1852